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M.S. Aerospace Engineering

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University of California, Los Angeles (UCLA)

M.S. Aerospace Engineering

GPA: 3.81 / 4.00

Graduated: **Fall 2025**

Thesis: Data-Driven Flight Dynamics Modeling for a Tailless Unmanned Aerial System (advisor: Kunihiro Taira)

B.S. Aerospace Engineering

GPA: 3.98 / 4.00, *Summa Cum Laude*

Graduated: **Fall 2024**

PROJECTS & EXPERIENCE

• Graduate Student Researcher | UCLA Taira Lab

Jan '25 – Dec '25

- Developed accurate flight dynamics models for Dihedral-90 (D90), a tailless unmanned aerial system (UAS), in collaboration with Prof. David Williams and PhD Candidate Sai Simon at the Illinois Institute of Technology.
- Applied Markov Chain Monte Carlo (MCMC) sampling through the Delayed Rejection Adaptive Metropolis (DRAM) algorithm to learn a set of stability derivatives that model significant dynamics.
- Explored the Metropolis-Hastings and Metropolis with Sampled Error Variance algorithms as alternative optimization approaches.
- Formed accurate flight dynamics models and identified dominant dynamics during takeoff and descent with Dutch roll flight segments.
- Performed five-fold cross-validation to mitigate overfitting.
- Performed Uncertainty Quantification (UQ) by constructing credible and prediction intervals on the final models to assess model accuracy.
- Analyzed parameter distributions to assess model and parameter sensitivity, and confidence in results.
- Explored the Morris screening method and Sobol indices as local and global parameter sensitivity metrics.
- Processed and performed all analysis on MATLAB, using Simulink to model the control systems on board D90 and to simulate flight dynamics.
- Ran 1000+ optimizations remotely on UCLA's Hoffman2 Cluster through the Linux command line.
- Developed a user-friendly MATLAB Graphic User Interface (GUI) to quickly and easily train and test sparse regression models from experimental flight data.
- The GUI allows the user to visually highlight significant dynamics from flight data and include various flight dynamics variables in the optimizations by comparing trends.
- The GUI trains models using the **Sparse Identification of Nonlinear Dynamics (SINDy)** algorithm, which performs least squares regression with an additional penalty on the size of parameters.
- Incorporated the option to run five-fold cross-validation to mitigate overfitting.
- Saved, compared, and analyzed models as sparsity was promoted and parameters were removed.
- Used the resulting models to seed the aforementioned MCMC optimizations.
- Thoroughly describe this research and the results in my Master's Thesis, seen [here](#).

• Undergraduate Research Assistant | UCLA Taira Lab

Jun '23 – Jan '25

- Researched and explored various optimization methods (e.g., Least Squares Regression, Sparse Identification of Nonlinear Dynamics, Least Absolute Shrinkage and Selection Operator) for modeling flight dynamics.
- Trained various models using each optimization method and tested them by comparing attitude and force simulation results with flight data from the D90 Tailless UAS.

- Performed all analysis on MATLAB, using Simulink to model the control systems on board D90 and to simulate flight dynamics.
- Laid the foundation for my graduate research by familiarizing myself with processing flight data and with optimization methods.
- Created and 3D-printed turbulent flow structures over straight and swept wings from computational fluid dynamics (CFD) simulation data.
- Wrote MATLAB scripts to triangulate a surface from CFD flow data, project color onto the surface indicating flow velocity, and export 3D-printable .stl, colored .obj, and .fbx files.
- Visualized the CFD simulation data and 3D-printable files using ParaView and Blender software, and removed any numerical discontinuities in the flow.
- Printed multiple prototype models, exploring the strength and complexity of the structures.
- Negotiated and received quotes from a multitude of outside vendors, eventually outsourcing to a professional resin printing company.
- Final 3D prints exceeded expectations, were given as gifts to lab donors, and are used to explain complex fluid mechanics topics.
- Additionally, I visited John. F. Kennedy High School in Granada Hills, CA, during an outreach event to help teach high school students about fluid mechanics.
- I designed and 3D-printed interactive gadgets and flow structures to make fluid mechanics engaging and more digestible for students.

• **Incompressible Lid-Drive Cavity Flow Solver with an Immersed Boundary Condition**

Mar '25 – Jun '25

- Wrote an incompressible lid-driven cavity flow solver using MATLAB and applied an immersed boundary condition.
- Explored the accuracy and stability properties of different finite-differencing and time-stepping methods (e.g., Explicit Euler, Runge-Kutta, Adams-Bashforth).
- Discretized the incompressible Navier-Stokes equations into discrete divergence, gradient, Laplacian, and nonlinear advection operators.
- Verified the code through spatial and temporal error convergence checks.
- Validated results by comparing u - and v -velocity profiles, vorticity contours, and streamlines at $Re = 400$ and $Re = 1000$ with results from a seminal incompressible flow solver paper.
- Implemented an immersed boundary condition to analyze flows around 2D structures. Demonstrated with a cylinder.
- Detailed this process in a report, seen [here](#).

• **The Silent Night Surveillant Aircraft Design Project**

Jan '25 – Mar '25

- Developed a full-scale aircraft for long-endurance surveillance missions.
- Designed the *Silent Night Surveillant* to launch from Santa's offshore *Christmas Carrier*, fly to the UCLA campus, identify and monitor a student of interest for 24 hours, and return to the ship for data processing and transmission back to the North Pole.
- Explored and compared different payloads and aircraft components to successfully carry out the mission.
- Sized the aircraft based on estimated total mass and mission specification (e.g., endurance, climb rate, stall speed, etc.) using fundamental flight dynamic equations.
- Seeded a Monte Carlo optimization approach with the preliminary sizing to refine the aircraft design.
- Accepted/rejected aircraft based on endurance, maximum climb angle, static margin, maximum velocity, and more.
- Iteratively adjusted seeding based on accepted aircraft configurations and reran Monte Carlo optimizations to achieve an optimized aircraft design.
- Lead the design and implementation of the guidance, navigation, and control (GNC) system, exploring different strategies (e.g., waypoint guidance, proportional navigation).
- Implemented logic to loiter around UCLA until the student was found, then circle the student for the mission duration.
- Used MATLAB's Simulink to model the control systems on board the aircraft and run simulations.
- Outlined this process in a report, seen [here](#).

• The *Mayfly* Spacecraft Design Project

Mar '24 – Jun '24

- Served as the lead launch vehicle and propulsion system engineer in the design of a full-scale spacecraft to perform a sample retrieval mission.
- Engineered the *Mayfly* spacecraft, designed to intercept and retrieve a sample from the 311P/PANSTARRS comet before returning back to Earth.
- Worked with the lead mission trajectory engineer and the systems engineer to ensure the spacecraft design adhered to the mission timeline.
- Collaborated closely with the power systems and thermal systems engineers to ensure sufficient power and adequate heat dissipation on board the spacecraft.
- Conducted trade studies between chemical and Solar Electric Propulsion (SEP) systems to determine the optimal propulsion system.
- Researched and evaluated multiple SEP systems using an iterative design approach to assess impacts on power consumption, spacecraft size and mass, and flight time.
- Assessed the Technology Readiness Level (TRL) of different SEP systems to reduce mission risks.
- Selected the high- I_{sp} variant of the NEXT 9a ion thruster ($I_{sp} = 4125\text{s}$) for its high efficiency, allowing for minimized propellant mass and maximized payload size.
- Reduced the thruster input power to 3kW to decrease the solar array size while still adhering to the mission timeline.
- Incorporated redundancy into the propulsion system design in case of mid-flight failures.
- Described this process in a report, seen [here](#).

• Autonomous Micro Air Vehicle Design Project

Mar '24 – Jun '24

- Designed and assembled an autonomous Micro Air Vehicle (MAV) capable of identifying the quickest path through a randomized set of waypoints.
- Lead engineer for the electrical subsystems on the drone, ensuring adequate power distribution and proper communication between components.
- Conducted research on electronic subsystems (e.g., LiDAR, flight computer hardware, motors, etc.) and chose components to meet mission requirements.
- Constructed the thrust profile of the motors, allowing for a current draw and power consumption estimates for the entire trajectory. Results were compared with the battery capacity to ensure adequate energy availability.
- Tested a variety of different frame materials through experiments and computational simulations.
- Soldered all components together, aligning the center of mass with the Inertial Measurement Unit (IMU).
- 3D modeled, printed, and tested various drone leg configurations for component placement and stability.
- Compared nearest neighbor, brute force, and k-opt waypoint sequencing methods to determine the optimal trajectory.
- Detailed this process in a report, seen [here](#).

• Independent RC Quadcopter Design Project

Jun '23 – Aug '23

- Designed, 3D-printed, and assembled an RC quadcopter from scratch.
- Sized drone according to 3D printer bed limitations and budget.
- Conducted copious amounts of research prior to designing and selecting each component of the drone.
- Rigorously selected motors and propellers in conjunction with a battery, after settling on a flight controller.
- Modeled the drone frame in SolidWorks and continuously 3D-printed prototypes, adjust the design with each iteration.
- Soldered all components together, added heat inserts to my 3D-printed frame, and assembled the drone.
- Learned how to configure the flight controller and adjust the drone for different flying styles.
- Continuing to redesign and experiment with new components on the drone to achieve smoothest flight.
- Studying to take HAM Amateur Radio License exam so I can attach a VTX chip and camera to the drone.

- **RC Airplane Design Project**

Jan '23 – Mar '23

- Lead a team of four students in designing, testing, and assembling a remote-controlled airplane.
- Improved communication and organizational skills, allowing my team to overcome a multitude of challenges.
- Consulted team members and set a goal of achieving the fastest plane award in the end competition.
- Constructed an Figure of Merit (FOM) chart to weigh the pros and cons of different component designs.
- Rigorously selected the airfoils for our wing and stabilizers based on lift and drag coefficients and the overall goals for the aircraft.
- Modeled entire plane in SolidWorks and 3D-printed engine mounts, stabilizer mounts, and landing gear mounts.
- Cut fuselage, wing, and stabilizers out of foam to reduce weight, inlaying a wooden spar to the wing.
- Iterated on component designed to achieve best performance.
- Placed in class a competition and won the award for the fastest and most maneuverable airplane.

- **Design/Build/Fly Manufacturing Team Member**

Jan '22 – Jun '22

- Joined the Manufacturing team and contributed to the design and assembly of a remote-controlled airplane placed in the AIAA Design/Build/Fly Competition.
- Helped manufacture and assemble the vertical and horizontal stabilizers of our plane.
- Learned how to run and analyze CFD simulations on different airfoils and wing designs using COMSOL.
- Manufactured and tested the vertical and horizontal stabilizers of the airplane.
- Placed 15th among 81 competing universities. At the time, this was the highest ever ranking in UCLA history.

- **Model Rocket Design Project**

Sept '22 – Dec '22

- Formed a team of students to design, assemble, test, and launch two model rockets.
- Modeled preliminary designs using the application OpenRocket and analyzed how different component designs affect apogee and speed.
- Conducted research on rocket design and communicated my understanding to my teammates.
- Held group meetings to discuss goals, divide up the work, and share research and ideas.
- Modeled individual components in SolidWorks and created an assembly design.
- Operated a laser cutter and 3D printer to manufacture components, and created a fiberglass layup body tube.
- Learned from failures after iteratively testing and redesigning components.
- Rockets exceeded expectations, with the second rocket achieving a ~ 3200 ft apogee.

- **Saddleback Robotics Osiris Mars Rover Team**

Jan '22 – Jun '22

- Joined the Chassis and Electrical teams and helped assemble a remote-controlled rover.
- Designed and 3D modeled inner components and electrical housing boxes within the rover's chassis.
- Soldered and assembled different circuitry within the rover.
- Helped in the repeated testing and redesign of the rover prior to competition.
- Entered the international University Rover Competition.

- **Misc. Projects / Work**

Jan '22 – Current

- Wrote a neural network in MATLAB that used the MNIST database to learn and identify hand written digits.
- Developed my personal website from scratch using HTML, CSS, and JavaScript.
- Have written multiple programs in C++ to calculate expenses, calculate income, play games, and more, sometimes utilizing Linux command line.
- Became a C++ class tutor after being recruited by my former professor. Held 3-hour drop-in office hour sessions to help students think through and correctly complete problems.
- Learned how to program an Arduino UNO and used it to display different light patterns.

CONFERENCE PRESENTATIONS

- **SoCal Fluids XVIII:** *Development of a Flight Dynamics Model for D90* – April 19, 2025 | University of Southern California

EXTRACURRICULAR ACTIVITIES & AWARDS

- **John F. Kennedy High School Outreach Event** – Explored fluid mechanics with high school students. Apr '25
- **Tau Sigma Academic Honor Society** – Member of academic honor society. May '23 – Current
- **Phi Theta Kappa Academic Honor Society** – Member of academic honor society. Apr '20 – Current
- **Saddleback College Tutor** – Computer science class tutor (C++), physics, chemistry, and math tutor. Aug '19 – May '22
- **Mentor Collective Trained Student Mentor** – Mentored STEM college students, helped identify interests. Jan '22 – Current